FYTA14

Hand in 5

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Question

A car with mass $M = 1000\,\mathrm{kg}$ moving at velocity $U_0 = 100\,\mathrm{km\,h^{-1}}$ suddenly hits a patch of ice and begins to slide. The total contact area between the wheels and the ice is $A = 3200\,\mathrm{cm^2}$, and it is observed to slide to a full stop in about $L = 300\,\mathrm{m}$.

(a) Calculate the thickness of the water layer and discuss whether it is a reasonable value.

(b) What is the time scale for stopping the car?

Additional help: Assume that the driver hits the breaks exactly at the same moment that the car hits the ice patch, so that the wheels are still.

(a)

To solve this problem we need to calculate the drag force, which is the force due to the no-slip condition in the thin water layer under the wheels. We state that that the thickness of the water layer is d, and the water has a viscosity of η, and the instantaneous velocity of the car is v, and that the velocity in the fluid is is $u(u_x(z), 0, 0)$, giving

$$\frac{d\vec{u}}{dt} + (\vec{u}\cdot\nabla)\,\vec{u} = \vec{g} - \frac{1}{\rho}\nabla p + \frac{1}{\rho}\eta\nabla^2\vec{v} + \frac{\varsigma + \frac{1}{3}\eta}{\rho}\nabla\,(\nabla\cdot\vec{u})\,. \tag{1}$$

We can say that the fluid is on a small scale in hydrostatic equilibrium $\vec{g} - \nabla p/\rho = 0$ and steady $d\vec{u}/dt = 0$. Because u_x is independent of x, we know that $\nabla\cdot\vec{u} = 0$. This gives the following equation:

$$0 = \eta\nabla^2\vec{v} = \eta\frac{d^2u_x}{dz^2}. \tag{2}$$

Solving this differential equation with the boundary condition that $u_x(z = d) = v$ and $u_x(z = 0) = 0$ gives:

$$u_x = \frac{v}{d}z. \tag{3}$$

This implies that the shear force and the total drag force is

$$\sigma_{xy} = \eta\frac{du_x}{dz} = \eta\frac{v}{d} \implies F_{\mathrm{drag}} = -\eta\frac{vA}{d}. \tag{4}$$

This drag force can also be described by newtons second law as

$$F_{\mathrm{drag}} = -\eta\frac{vA}{d} = M\frac{dv}{dt} \implies \frac{dv}{dt} = -\eta\frac{vA}{Md}. \tag{5}$$

This is a differential equation which can be solved by using the separation of variables as:

$$\frac{dv(t)}{v(t)} = -\frac{\eta A}{Md}dt \implies \int\frac{dv(t)}{v(t)} = -\int\frac{\eta A}{Md}dt \tag{6}$$

$$\implies \ln|v(t)| = -\frac{\eta A}{Md}t + C \implies v(t) = U_0e^{-\frac{\eta A}{Md}t}. \tag{7}$$

We have now obtained an expression for the velocity through time. If we integrate this from 0 to infinity, we should obtain the stopping distance as

$$L = \int_0^\infty U_0e^{-\frac{\eta A}{Md}t}dt = \left[-U_0\frac{Md}{\eta A}e^{-\frac{\eta A}{Md}t}\right]_0^\infty = U_0\frac{Md}{\eta A} \implies d = \frac{L\eta A}{U_0M}. \tag{8}$$

Answer

Using the vales $\eta_{\mathrm{water}} = 1.0\,\mathrm{mPa\,s}$, $M = 1000\,\mathrm{kg}$, $U_0 = 100\,\mathrm{km\,h^{-1}}$, $A = 3200\,\mathrm{cm^2}$, and $L = 300\,\mathrm{m}$ we obtain $d = 3.46\,\mu\mathrm{m}$. This value is reasonable although the thickness is very small.

(b)

In the second part we are asked on what timescale the car is stopped. This is however a difficult task since the velocity is exponentially declining, implying that the car dos'nt theoretically ever stop! However, we can define a lower limit, where at that speed we define that the car has stopped.

We can define the car to have stopped when the speed is $v(t) < 0.1\,\text{meter/s}$. This gives us

$$0.1 < U_0 e^{-\frac{\eta A}{dM} t_{\text{stop}}}, \tag{9}$$

which can be solved numerically. Solving for t_{stop} gives us:

$$t_{\text{stop}} < \frac{dM}{\eta A} \ln\left(\frac{U_0}{0.1\,\text{m}\,\text{s}^{-1}}\right) = \frac{3.46 \times 10^{-6}\,\text{m} \cdot 1000\,\text{kg}}{1 \times 10^{-3}\,\text{Pa} \cdot 0.32\,\text{m}^2} \ln\left(\frac{27.78\,\text{m}\,\text{s}^{-1}}{0.1\,\text{m}\,\text{s}^{-1}}\right) \approx 61\,\text{s}. \tag{10}$$

Answer

Theoretically the car newer stops, but using approximations we can conclude that in reality the car stops after one minute.